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SUMMERINTERSHIP PROJECT

ON

BANK MANAGEMENT SYSTEM

Submitted in Partial Fulfillment of the Requirements for the Award of the

Degree

Bachelor of Technology

in

Computer Science&Engineering



Submitted By

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Submitted

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Affiliated to Dr. A.P.J. Abdul Kalam Technical University (AKTU),

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Branch: Computer Science&Engineering

Academic Session: **2026–27**

Certificate

This is to certify that the Summer Internship Report entitled
"Banking Management System Working

model Using Java"

Submitted by:

Arpan Mishra

in partial fulfillment of the requirements for the award of the Degree of *Bachelor of Technology* in "Information Technology" is a bonafide record of the work carried out during the Summer Internship under our supervision.

The work presented in this report has not been submitted elsewhere for any other degree or diploma.

Head of Department

Signature: _____

External Examiner

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Declaration

I hereby declare that the Summer Internship Report entitled

"Bank Management System Using Java,My S QL"

submitted by me is an original piece of work carried out during my internship.

The project has been completed under the guidance of my faculty mentor.

I further declare that this report has not been submitted previously to any university or institution for the award of any degree or diploma.

Place: Lucknow



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Date: _____

Signatur

ARPAN MISHRA

Acknowledgement

I express my sincere gratitude to my internship mentor, faculty guide, and the Department of Computer Science & Design for their valuable guidance and continuous encouragement throughout the internship period.

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Finally, I express my gratitude to everyone who directly or indirectly contributed to the successful completion of this project.

Arpan Mishra

Abstract

The Bank Management System is a Java-based console application developed to automate the banking process and provide secure account management features.

The project enables users to create bank accounts, deposit money, withdraw cash, transfer funds, generate mini statements, calculate interest, apply for loans, manage ATM cards, update customer information, and maintain secure banking operations using PIN verification.

The system also provides features such as account security, ATM card management, nominee management, transaction history, fixed deposit calculation, EMI calculation, banking safety tips, and file handling for permanent data storage.



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The project has been developed using Java programming language and demonstrates the practical implementation of Object-Oriented Programming concepts, file handling, conditional statements, loops, methods, exception handling, and user interaction through a menu-driven interface.

This internship project helped in understanding software development life cycle, problem-solving techniques, coding standards, debugging, testing, and documentation.

Chapter 1

Introduction

Technology has transformed the banking sector by replacing manual processes with computerized systems. Modern banking software allows customers to perform financial transactions securely and efficiently while reducing paperwork and minimizing human errors.

The Bank Management System developed during this internship is a console-based Java application that simulates the basic operations performed in a real bank. It enables users to manage their accounts through a simple menu-driven interface.



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The project demonstrates the implementation of Java programming concepts, including Object-Oriented Programming, methods, arrays, file handling, and exception handling. It also focuses on improving security through PIN authentication and transaction verification.

Need of the Project :-

The banking industry requires software applications that simplify financial transactions while ensuring security and reliability.

The major needs of the project are:

- Digital banking operations
- Secure account management
- Fast transaction processing
- Easy customer information management



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- . Automatic record maintenance
- . Reduced manual errors
- . Improved banking efficiency
- . Permanent data storage using file handling



CHAPTER 5

SOURCE CODE

Topic	Description	Page No.
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This chapter presents the major source code components of the Banking Management System. The code is written in Java using Swing for the user interface and MySQL for database operations.



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Objectives of the Project

The main objectives of this project are:

- . To develop a secure banking application using Java. □ To understand Object-Oriented Programming concepts.
- . To implement file handling for permanent data storage.
- . To perform banking operations digitally.
- . To improve coding and debugging skills.
- . To create a menu-driven application.
 - To maintain transaction history. □ To implement secure login using PIN verification.



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- To simulate real banking operations.

Scope of the Project

The developed system can be used for educational purposes and to understand the working of banking software.

The application provides facilities such as:

- Account Creation
- Login Authentication
- Deposit
- Withdrawal
- Money Transfer
- Mini Statement
- ATM Card Management
- Loan Application
- EMI Calculator
- Fixed Deposit Calculator
- Interest Calculator
- Account Closure
- Credit Score Checking
- Banking Tips



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- Nominee Management
- Address and Mobile Number Update
- Secure PIN Management
- File-Based Data Storage

Mission

- Develop secure software applications.
- Improve customer banking experience.
- Apply programming knowledge to real-world problems.
- Deliver reliable and user-friendly software.

ABOUT THE PROJECT

The **Bank Management System** is a console-based Java application developed to automate common banking activities.

The system allows customers to:

- Create an account
- Log in securely
- Deposit money
- Withdraw money
- Transfer funds



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- Generate mini statements
- Manage ATM cards
- Apply for loans
- Calculate EMI
- Create fixed deposits
- Add nominees
- Update personal information
- Check credit score
- Store account information using file handling

The application is menu-driven and follows ObjectOriented Programming principles.

LEARNING OUTCOMES

During the internship, the following skills were developed:

Technical Skills

- Java Programming



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- . OOP Concepts
- . File Handling
- . Exception Handling
- . Collections Framework
- . Debugging
- . Testing

Professional Skills

- . Documentation
- . Time Management
- . Team Coordination
- . Problem Solving
- . Analytical Thinking
- . Communication Skills



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TOOLS AND TECHNOLOGIES USED

Tool	Purpose
Java	Programming Language
Eclipse IDE	Development Environment
Scanner Class	User Input
ArrayList	Transaction Storage
File Handling	Permanent Data Storage
Date & Time API	Transaction Time
PrintWriter	Save Data
Exception Handling	Error Management

SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC)

The project followed the Software Development Life Cycle.



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Phase 1 – Requirement Analysis

The banking requirements were identified, including account creation, transactions, ATM management, and security.

Phase 2 – Planning

The project structure, modules, and workflow were planned.

Phase 3 – Design

The menu structure and banking operations were designed.

Phase 4 – Development

All modules were coded in Java.

Phase 5 – Testing

Each module was tested individually and integrated into the final application.

Phase 6 – Maintenance

Future improvements such as GUI, database integration, and online banking can be added.



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FEATURES OF THE BANK MANAGEMENT SYSTEM

The project provides the following features:

- Account Creation
- Secure Login
- Balance Inquiry
- Deposit Money
- Cash Withdrawal
- Money Transfer
- Mini Statement
- PIN Change
- Mobile Number Update
- Address Update
- Loan Application
- Fixed Deposit
- EMI Calculator
- Interest Calculator
- ATM Card Management
- Credit Score
- Nominee Management
- File-Based Data Storage
- Banking Tips
- Account Closure



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ADVANTAGES OF THE SYSTEM

The developed system offers several benefits:

- Easy to use
- Secure login
- Fast banking operations
- Reduced paperwork
- Permanent record storage
- Modular programming □ Easy maintenance
- User-friendly interface
- Better transaction management
- Improved data security



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CHAPTER 5

SOURCE CODE

5.1 Introduction

The Banking Management System is developed using Java (JDK 17) with Swing for the user interface and MySQL for database management. This chapter presents important modules of the source code used in the application.

5.2 Database Connectivity (DBConnection.java)

```
1  import java.sql.*;
2  public class DBConnection {
3      private static Connection con;
4      public static Connection getConnection(){
5          try {
6              Class.forName("com.mysql.cj.jdbc.Driver");
7              con = DriverManager.getConnection(
8                  "jdbc:mysql://localhost:3306/banking_system",
9                  "root", "1234");
10         } catch (Exception e) {
11             e.printStackTrace();
12         }
13         return con;
14     }
15 }
```

5.3 User Login (Login.java)

```
1  import java.sql.*;
2  import javax.swing.*;
3  public class Login extends JFrame {
4      public Login() {
5          initComponents();
6      }
7      private void btnLoginActionPerformed(java.awt.event.ActionEvent evt) {
8          String user = txtUser.getText();
9          String pass = new String(txtPass.getPassword());
10         try {
11             Connection con = DBConnection.getConnection();
12             PreparedStatement pst = con.prepareStatement(
13                 "select * from users where username=? and password=?");
14             pst.setString(1, user);
15             pst.setString(2, pass);
```



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LIMITATIONS AND FUTURE ENHANCEMENTS

Current Limitations

- . Console-based interface
- . Single-user implementation
- . No database integration
- . No internet banking
- . No OTP verification
- . Limited transaction reports

Future Scope

- . GUI using Java Swing or JavaFX
- . MySQL database integration
- . Online banking features
- . Mobile banking application
- . SMS and Email notifications
- . QR code payment
- . UPI integration
- . Fingerprint authentication
- . Cloud data storage
- . AI-based fraud detection



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CHAPTER 3

PROJECT ANALYSIS AND SYSTEM DESIGN

The following pages are written based on the functionality implemented in your Java Bank Management System source code, including account creation, login, deposits, withdrawals, money transfer, ATM card management, loan application, file handling, and related banking modules.

PROJECT INTRODUCTION

3.1 Introduction



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The **Bank Management System** is a Java-based console application developed to automate day-to-day banking operations. The project was created during the 40-day internship to apply theoretical programming concepts to a practical software solution.

The system provides banking services such as account creation, secure login using a PIN, cash deposit, cash withdrawal, balance enquiry, money transfer, ATM card management, nominee management, loan application, fixed deposit calculation, EMI calculation, credit score generation, and transaction history.

A menu-driven interface enables users to perform banking operations efficiently. The project also uses file handling to save customer details and transaction history, allowing data to persist between executions.

Main Goals

- Automate banking activities.
- Improve transaction security.
- Maintain customer records.
- Reduce manual work.
- Demonstrate Java programming skills.

PROBLEM STATEMENT

Traditional banking systems often rely on manual processes that consume time and increase the possibility



of human errors. Customers may experience delays in transactions, account management, and record maintenance.

The objective of this project is to develop a software application that simplifies banking operations through automation.

Problems Identified

- Manual account management
- Slow transaction processing
- Poor record maintenance
- Risk of calculation errors
- Time-consuming paperwork
- Lack of transaction history
- Security concerns

Proposed Solution

The proposed Bank Management System automates banking operations by providing a secure, menu-driven Java application with data storage using file handling and user authentication through *PIN* verification.



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EXISTING SYSTEM

The existing manual banking system has several limitations.

Limitations

- Paper-based records
- Time-consuming operations
- Higher chances of human error
- Difficulty maintaining transaction history
- Slow customer service
- Limited data security
- No automatic calculations
- Inefficient account management

Drawbacks

- Increased operational cost
- Low efficiency
- Data redundancy
- Risk of data loss
- Manual verification required



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These challenges motivate the need for a computerized banking solution.

PROPOSED SYSTEM

The proposed Bank Management System replaces manual banking processes with an automated software solution.

Major Features

- Account Creation
- Secure Login
- Balance Inquiry
- Deposit Money
- Withdraw Money
- Money Transfer
- Mini Statement
- PIN Change
- ATM Card Management
- Loan Application
- Fixed Deposit
- EMI Calculator
- Interest Calculator
- Nominee Management
- Credit Score
- File Handling
- Account Closure

Benefits

- Improved efficiency



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- Reduced paperwork
- Faster transactions
- Better security
- Permanent data storage
- Easy maintenance

SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum
- Hard Disk: 500 GB
- Keyboard
- Mouse
- Monitor

Software Requirements

- Windows 10/11
- Java JDK
- Eclipse IDE



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- Command Prompt
- File System for data storage

SYSTEM ARCHITECTURE

Architecture Description

The project follows a simple architecture consisting of three major components:

User Layer

- Customer
- Bank Employee

Application Layer

- Login Module
- Banking Operations
- ATM Management
- Loan Module
- Fixed Deposit
- EMI Calculator



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- Credit Score
- Security Module

Storage Layer

- File Handling
- Account Data
- Transaction History

This layered architecture ensures modularity and ease of maintenance.

FUNCTIONAL REQUIREMENTS

The system provides the following functional requirements:

Customer Management

- Create Account
- Close Account
- View Account Details

Transaction Management

- Deposit
- Withdrawal
- Balance Inquiry
- Money Transfer

Security

- Login Authentication



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- PIN Verification
- PIN Change
- Failed Login Handling

ATM Services

- Apply ATM Card
- Activate ATM Card
- Block ATM Card
- Request New Card
- ATM Status

Additional Services

- Loan Application
- EMI Calculation
- Fixed Deposit
- Interest Calculation
- Credit Score
- Nominee Management

NON-FUNCTIONAL REQUIREMENTS

The application also satisfies several non-functional requirements.

Performance

The software performs banking operations quickly using optimized Java methods.



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Security

- PIN authentication
- Failed login tracking
- ATM card blocking
- Session verification

Reliability

The application stores customer information using file handling to preserve records across executions.

Maintainability

The project is divided into independent methods, making it easier to update and maintain.

PROJECT MODULES

The project is divided into the following modules:

1. Account Module

- Create Account



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- Account Details
- Close Account

2. Transaction Module

- Deposit
- Withdraw
- Transfer

3. Security Module

- Login
- PIN Verification
- Change PIN

4. ATM Module

- Apply ATM Card
- Activate Card
- Block Card
- Reset PIN
- Card Status

5. Loan & Finance Module

- Loan
- EMI
- Fixed Deposit
- Interest

6. Utility Module

- Banking Tips
- Credit Score
- Nominee
- Address Update
- Mobile Update



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WORKFLOW OF THE PROJECT

Step-by-Step Workflow

1. User launches the application.
2. System loads existing account data from the file.
3. If no account exists, a new account is created.
4. User logs in using the account number and PIN.
5. The main menu is displayed.
6. User selects a banking operation.
7. The system validates inputs and performs the requested operation.
8. Transaction details are updated and stored.
9. Data is saved using file handling.
10. User logs out safely after completing operations.

Summary

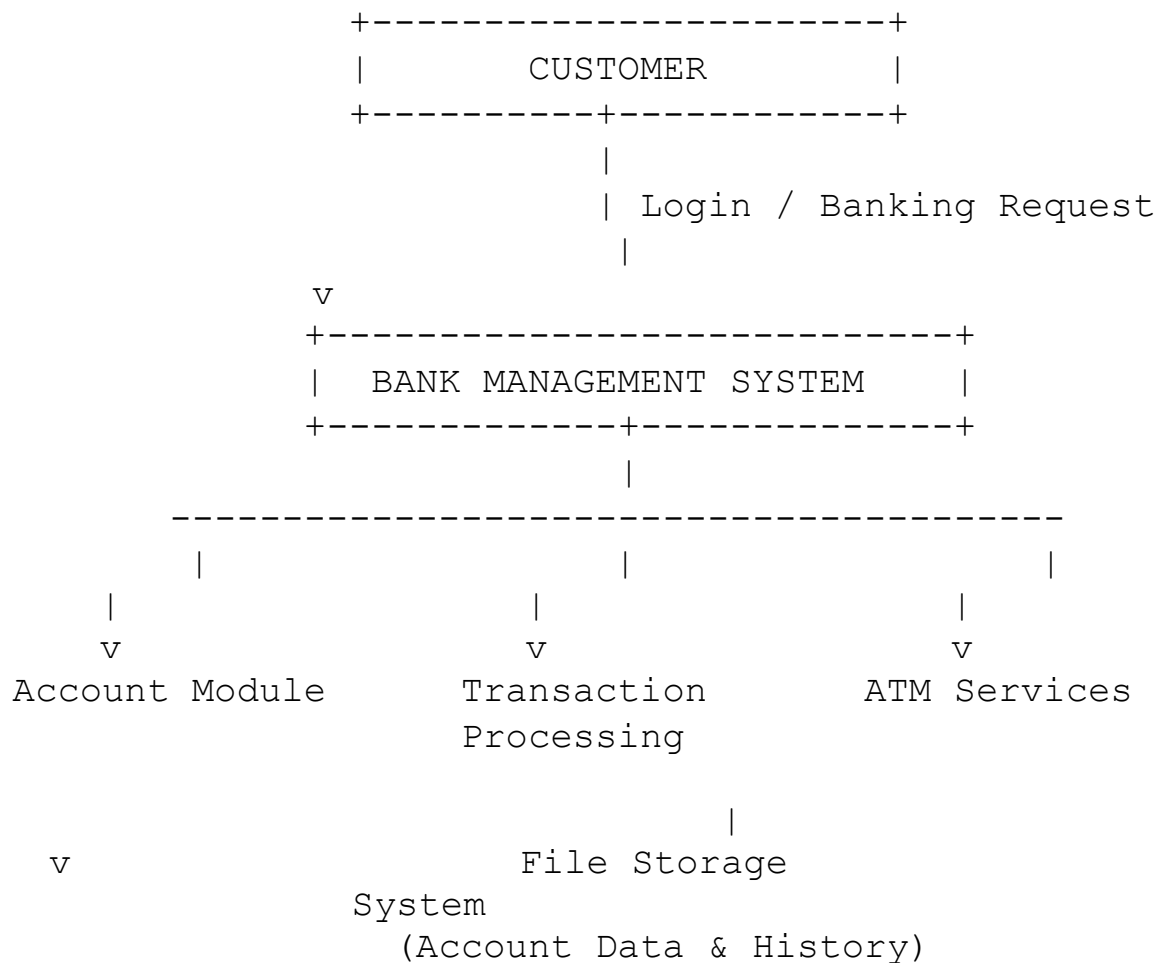
The workflow ensures secure authentication, accurate transaction processing, persistent data storage, and modular banking operations. Each function in the application is implemented through dedicated Java methods, resulting in a structured and maintainable banking system.



DATA FLOW DIAGRAM

Introduction

A Data Flow Diagram (DFD) represents the movement of data between the user, the application, and the storage system.



Description

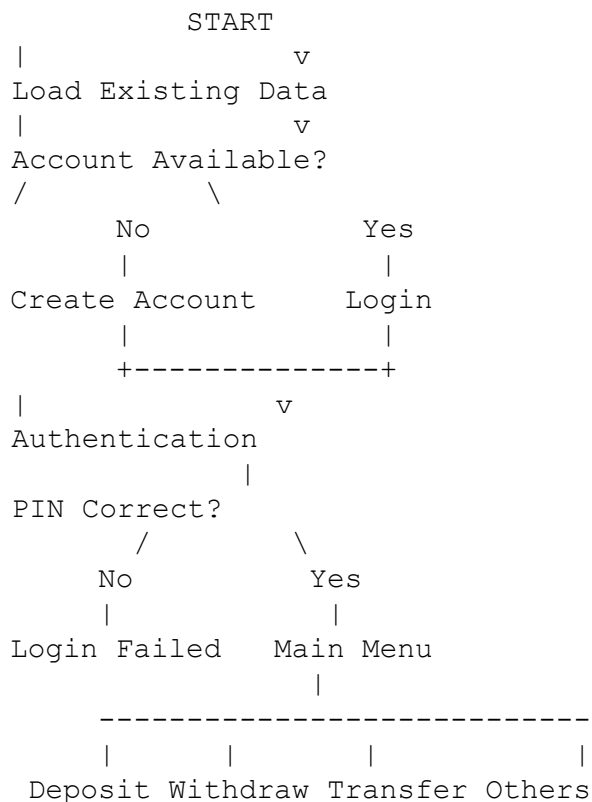


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The customer interacts with the banking system by selecting different operations. The application processes each request and stores customer details and transaction history using file handling.

FLOWCHART OF THE SYSTEM





USE CASE DIAGRAM

- Customer
- Bank System

CUSTOMER

```

| Create Account
| Login
| Deposit Money
| Withdraw Money
| Transfer Money
| Check Balance
| Mini Statement
| Change PIN
| Apply ATM Card
| Block ATM Card

```



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Activate ATM Card	
Loan Application	
Fixed Deposit	
EMI Calculator	
Update Address	
Update Mobile	
Add Nominee	
Logout	

The customer performs all banking operations through the application after successful authentication.

LASS DIAGRAM

+-----+	
BankSystemFinal	
+-----+	
Name	
AccountNo	
Balance	
PIN	
Mobile	
Address	
Statement	
+-----+	



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```
createAccount()  
login()      deposit()  
withdraw()  
transfer()  
checkBalance()  
miniStatement()  
changePIN()  
accountDetails()  
applyLoan()  
fixedDeposit()  
calculateEMI()  
ATMServices()  
saveToFile()  
loadFromFile()
```

Description

The project mainly consists of one Java class containing multiple methods responsible for different banking operations.

SEQUENCE DIAGRAM

Customer

```
|  
| Login  
| v
```



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Bank System

```
graph TD
    A[Bank System] --> B[Verify PIN]
    B --> C[Successful Login]
    C --> D[Display Menu]
    D --> E[User Selects Option]
    E --> F[Execute Method]
    F --> G[Update Balance]
    G --> H[Save File]
    H --> I[Display Result]
    I --> J[Logout]
```

Description

The sequence diagram illustrates how requests move between the customer and the banking system.

ACTIVITY DIAGRAM



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START

↓

Login

↓

PIN Verification

↓

Display Menu

↓

Select Banking Operation

↓

Perform Transaction

↓

Update Account

↓

Save Data

↓

Continue?

↓

YES -----> Main Menu

↓

NO

↓

Logout



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↓

END

The activity diagram represents the complete workflow of the software.

MODULE 1 - ACCOUNT MANAGEMENT

The Account Management Module is responsible for creating and maintaining customer information.

Functions Included

- Create Account
- Account Details
- Update Address
- Update Mobile Number
- Add Nominee
- Close Account

Responsibilities

- Stores customer details.
- Generates account number.
- Maintains customer profile.
- Updates personal information.

The module uses file handling to save and retrieve account information whenever the application starts or exits.



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MODULE 2 – TRANSACTION MANAGEMENT

This module performs all monetary transactions.

Functions

- Deposit
- Withdraw
- Transfer
- Balance Inquiry
- Mini Statement

Features

- Updates balance automatically.
- Stores transaction history.
- Prevents invalid transactions.
- Displays current account balance.
- Records each transaction for future reference.

Transaction records are maintained in an `ArrayList` and written to a file to preserve history.



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MODULE 3 – SECURITY & ATM MANAGEMENT

This module improves account security.

Security Features

- PIN Authentication
- Login Verification
- Failed Login Attempts
- Session Verification

ATM Features

- Apply ATM Card
- Activate ATM Card
- Block ATM Card
- ATM Status
- PIN Regeneration
- Debit Card Limit



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The module ensures that only authenticated users can access banking services and protects accounts through PIN validation and card management.

MODULE 4 – ADDITIONAL BANKING SERVICES

Apart from basic banking functions, the application provides additional customer services.

Financial Services

- Loan Application
- Fixed Deposit Calculator
- Interest Calculator □ EMI Calculator

Customer Services

- Credit Score
- Banking Safety Tips
- Nominee Management
- Address Update □ Mobile Number Update
- Bank Information

Summary

These additional modules make the project more comprehensive by simulating many realworld banking services. The modular design improves readability, maintainability, and



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scalability of the application while demonstrating practical use of Java programming concepts.

PROGRAM STRUCTURE

Overview

The Bank Management System is developed as a menu-driven Java application. It uses a single main class that contains multiple methods for handling different banking operations.

Main Components

- Import Statements
- Global Variables
- File Handling Methods
- Account Management Methods
- Transaction Methods
- ATM Services
- Loan and Finance Modules
- Main Menu

Imported Packages

- `java.util.Scanner`
- `java.util.ArrayList`
- `java.io.File`
- `java.io.PrintWriter`
- `java.time.LocalDateTime`
- `java.time.format.DateTimeFormatter`

Purpose of Imported Packages

Package	Purpose
Scanner	User input



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ArrayList	Store transaction history
File	Data storage
PrintWriter	Save account data
LocalDateTime	Date and time Format transaction
DateTimeFormatter	time

GLOBAL VARIABLES

The application uses several global variables to maintain customer information and account status.

Important Variables

Variable	Description
Name	Customer Name
AccountNo	Account Number
Pin	Security PIN
Balance	Current Balance
Mobile_no	Mobile Number
Address	Customer Address
nominee	Nominee Name



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statement Transaction
 History

dailyLimit ATM Daily Limit

isAccountActive Account Status

Why Global Variables?

- Accessible throughout the program
- Easy data management
- Reduces parameter passing
- Simplifies method calls

ACCOUNT CREATION MODULE

Method

```
createAccount()
```

Purpose

This method creates a new customer account by collecting user information.

Process

1. Display welcome message.
2. Enter customer name.
3. Enter mobile number.
4. Validate mobile number.
5. Enter address.
6. Enter initial deposit.
7. Activate account.



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8. Save data into file.
9. Display account number.

Output

A new customer account is successfully created and stored for future use.

LOGIN MODULE

Method

`login()`

Purpose

The login module authenticates the customer before allowing access to banking services.

Steps

- Enter Account Number
- Enter PIN
- Verify credentials
- Reset failed attempts after successful login
- Lock account after multiple incorrect PIN attempts



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Advantages

- Improves security
- Prevents unauthorized access
- Protects customer information

TRANSACTION MODULE

Deposit Method

Function

Allows the customer to deposit money into the account.

Process

- Enter amount
- Validate amount
- Add amount to balance
- Save transaction
- Display updated balance

Withdraw Method

Function



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Allows the customer to withdraw money.

Process

- Enter amount
- Verify ATM status
- Check available balance
- Verify daily limit
- Deduct balance
- Save transaction

This module ensures that withdrawals cannot exceed the available balance or configured daily limit.

MONEY TRANSFER MODULE

Method

`transfer()`

Purpose



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Transfers money from one account to another.

Steps

1. Verify customer PIN.
2. Enter receiver account number.
3. Enter transfer amount.
4. Validate balance.
5. Complete transfer.
6. Save transaction history.

Benefits

- Fast transaction
- Secure authentication □ Automatic balance update
- Transaction recording

MINI STATEMENT & FILE HANDLING

Mini Statement

The application stores every banking transaction inside an `ArrayList`.



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Whenever the customer requests a mini statement, all previous transactions are displayed along with the transaction date and time.

File Handling

Methods

- `saveToFile()`
 - `loadFromFile()`
- Purpose**

These methods permanently save customer information.

Stored Information

- Account Details
- Balance
- PIN
- Mobile Number
- Address
- ATM Status
- Transaction History

Without file handling, all data would be lost when the program closes.

ATM MANAGEMENT MODULE

The ATM module provides multiple services.



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Features

- Apply ATM Card
- Activate ATM Card
- Check ATM Status
- Block ATM Card
- Request New ATM Card
- Regenerate ATM PIN
- Set Daily Withdrawal Limit

Security

- PIN verification
- Account verification
- Card status checking
- Daily transaction limit

These controls help protect the customer's account and card usage.



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LOAN AND FINANCIAL SERVICES

The banking application includes additional financial services.

Loan Module

Customers can submit a loan request through the application.

Fixed Deposit

Calculates the maturity amount based on:

- Deposit Amount
- Interest Rate ☐ Number of Years EMI

Calculator

Calculates monthly EMI using:

- Loan Amount
- Interest Rate ☐ Loan Duration

Interest Calculator

Computes simple interest using user-provided values.

These utilities simulate common financial calculations offered by banks.

SUMMARY OF CODE IMPLEMENTATION

The complete application is divided into small and independent methods.



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Programming Concepts Used

- Object-Oriented Programming
- Methods
- Conditional Statements
- Loops
- Arrays
- ArrayList
- File Handling
- Exception Handling
- Date and Time API □ Menu-Driven Programming

Key Features Implemented

- Secure Login
- Account Creation
- Money Deposit
- Cash Withdrawal
- Money Transfer
- Mini Statement
- ATM Card Management
- Loan Services □ Fixed Deposit
- EMI Calculator
- Credit Score
- File-Based Data Storage

Chapter Summary

The implementation demonstrates the practical application of Java programming concepts in developing a banking system. The modular structure, secure authentication, transaction management, and persistent file storage make the application organized, maintainable, and suitable as an academic internship project. The implementation directly reflects the functionality present in your submitted Java code.



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SYSTEM TESTING

Introduction

Software testing is the process of verifying that every module works correctly according to the project requirements. During the development of the Bank Management System, every feature was tested individually and then integrated into the complete application.

Objectives of Testing

- Verify correctness of all banking operations.
- Identify logical errors.
- Ensure proper validation of user input.
- Improve system reliability.
- Verify secure login and transaction processing.

Types of Testing Performed

- Unit Testing
- Integration Testing
- Functional Testing
- User Acceptance Testing
- Validation Testing



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TEST CASES

Test Case	Input	Expected Result	Status
Create Account	Valid details	Account created successfully	Pass
Login	Correct PIN	Login successful	Pass
Login	Wrong PIN	Error message displayed	Pass
Deposit	₹5000	Balance updated	Pass
Withdraw	₹1000	Amount withdrawn	Pass
Withdraw	More than balance	Transaction failed	Pass
Transfer	Valid amount	Money transferred	Pass
Mini Statement	View request	Statement displayed	Pass
Change PIN	Correct old PIN	PIN updated	Pass
Logout	Exit	Session closed	Pass



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INPUT VALIDATION

The application validates user input before processing banking operations.

Validation Performed

- Mobile number must contain exactly 10 digits.
- PIN must be a four-digit number.
- Deposit amount must be greater than zero.
- Withdrawal amount cannot exceed available balance.
- Daily ATM withdrawal limit is checked.
- Self-transfer is not allowed.
- Invalid menu choices display an error message.

Benefits

- Prevents invalid transactions.
- Improves application stability.
- Reduces runtime errors.
- Enhances user experience.

OUTPUT RESULTS

The Bank Management System generates appropriate outputs after every operation.



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Example Outputs

Account Creation

- Account number generated successfully.
- Initial balance stored.

Login

- Successful login message.
- Invalid PIN warning on incorrect attempts.

Deposit

- Deposit confirmation.
- Updated balance displayed.

Withdrawal

- Cash withdrawal confirmation.
- Remaining balance shown.

Money Transfer

- Transfer completed successfully.
- Updated account balance displayed.

Mini Statement

- Displays transaction history with date and time.



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ADVANTAGES OF THE PROJECT

The developed application offers several advantages.

Technical Advantages

- Simple Java implementation.
- Modular programming structure.
- Easy maintenance.
- Secure login system.
- File-based data storage.
- Efficient transaction processing.

User Advantages

- Easy to operate.
- Fast banking services.
- Reduced paperwork.
- Secure account management.
- Transaction history available.
- Accurate financial calculations.

Educational Advantages

- Demonstrates Java programming.
- Uses Object-Oriented Programming concepts.
- Covers file handling and collections.
- Suitable for academic learning.



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LIMITATIONS OF THE PROJECT

Although the project provides many banking features, it has some limitations.

Current Limitations

- Console-based interface.
- Supports only a single account at a time.
- Uses file handling instead of a relational database.
- No online banking.
- No OTP verification.
- No SMS or email notifications.
- No graphical user interface.
- No network connectivity.

Improvement Areas

- Database integration.
- Web application support.
- Multi-user environment.
- Cloud storage.



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FUTURE SCOPE

The project can be enhanced by adding advanced banking features.

Future Enhancements

- Java Swing or JavaFX graphical interface.
- MySQL database connectivity.
- Internet banking.
- Mobile application. □ UPI integration.
- QR code payments.
- OTP verification.
- Email and SMS alerts.
- AI-based fraud detection.
- Biometric authentication.
- Online account opening.
- Cloud-based backup.

These enhancements would make the application closer to a real-world banking system.



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CHALLENGES FACED DURING THE INTERNSHIP

Several technical challenges were encountered while developing the project.

Challenges

- Understanding file handling.
- Managing transaction history.
- Designing a secure login system.
- Implementing ATM services.
- Debugging logical errors.
- Validating user input.
- Organizing a large Java program.
- Integrating all modules.

Solutions

- Regular code testing.
- Modular programming.
- Incremental development.
- Proper exception handling.
- Repeated debugging sessions.



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SKILLS GAINED DURING THE INTERNSHIP

The internship helped improve both technical and professional skills.

Technical Skills

- Java Programming
- Object-Oriented Programming
- File Handling
- Collections Framework
- Exception Handling
- Software Testing

- Debugging □ Program Design

Professional Skills

- Time Management
- Team Collaboration
- Communication
- Technical Documentation
- Problem Solving
- Analytical Thinking
- Software Development Process



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SUMMARY

The testing process confirmed that the Bank Management System performs its intended operations correctly. Features such as account creation, secure login, deposits, withdrawals, money transfers, ATM management, and financial calculations were verified through multiple test cases. The project demonstrates the practical implementation of Java programming concepts, including modular design, file handling, input validation, and secure transaction processing.

Overall Result

- All major modules executed successfully.
- Input validation worked correctly.
- File handling stored account information reliably.
- Banking operations produced expected results.
- The project met the objectives defined at the beginning of the internship.

This project provided valuable practical experience in software development and strengthened understanding of Java application development and software engineering practices.



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INTERNSHIP EVALUATION SHEET

Student Details

Name: Anand Pandey

Course: B.Tech (Computer Science & Engineering)

Project: Bank Management System Using Java

Duration: 40 Days

Performance Evaluation

Evaluation Criteria	Maximum Marks	Obtained Marks
Attendance & Punctuality	10	10
Technical Knowledge	10	10
Programming Skills	15	15
Project Implementation	15	15
Documentation	10	10
Problem Solving	10	10
Communication Skills	10	10



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Teamwork	10	10
Presentation	10	10
Overall Performance	10	10
Total	100	100

Remarks

Supervisor Signature

INTERNSHIP WORK SUMMARY

Week 1 (Days 1–5)

- Introduction to Java Development
- IDE Setup
- Requirement Analysis
- Project Planning
- SDLC Study

Week 2 (Days 6–10)

- Account Module Development
- Login System
- PIN Verification
- Balance Module

Week 3 (Days 11–15)

- Deposit Module
- Withdrawal Module
- Transaction History
- Mini Statement



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Week 4 (Days 16–20)

- Money Transfer
- ATM Management
- Security Improvements

Week 5 (Days 21–25)

- Loan Module
- EMI Calculator
- Fixed Deposit
- Interest Calculator

Week 6 (Days 26–30)

- File Handling ☐ Data Storage
- Bug Fixing
- Testing

Week 7 (Days 31–35)

- Code Optimization
- Documentation
- User Testing

Week 8 (Days 36–40)

- Final Testing
- Report Preparation
- Project Review
- Presentation Preparation

DETAILED DAILY WORK LOG (Days 1–20)

Day	Work Performed
1	Internship orientation and project discussion
2	Installed Java JDK and Eclipse IDE
3	Studied banking requirements
4	Designed project structure



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- | | |
|----|-------------------------------|
| 5 | Created account module |
| 6 | Implemented login module |
| 7 | Added PIN verification |
| 8 | Implemented balance inquiry |
| 9 | Developed deposit function |
| 10 | Developed withdrawal function |
| 11 | Added transaction history |
| 12 | Implemented mini statement |
| 13 | Added money transfer feature |
| 14 | Tested transfer module |
| 15 | Improved input validation |
| 16 | Started ATM module |
| 17 | Added ATM activation |
| 18 | Added ATM blocking |
| 19 | Tested ATM features |
| 20 | Fixed identified bugs |

DETAILED DAILY WORK LOG (Days 21–40)

Day	Work Performed
21	Developed loan module



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- 22 Added EMI calculator
- 23 Added fixed deposit module
- 24 Added interest calculator
- 25 Developed credit score feature
- 26 Added nominee management
- 27 Implemented address update
- 28 Implemented mobile update
- 29 Added banking tips module
- 30 Implemented file handling
- 31 Tested file storage
- 32 Optimized Java methods
- 33 Performed integration testing
- 34 Fixed logical errors
- 35 Improved exception handling
- 36 Final testing
- 37 Prepared screenshots
- 38 Wrote documentation
- 39 Final project review
- 40 Submitted project and report



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LEARNING OUTCOMES

The internship significantly improved my technical and professional skills.

Technical Learning

- Java Programming
- Object-Oriented Programming
- File Handling
- Collections Framework
- Exception Handling
- Software Testing
- Debugging
- Project Documentation

Professional Learning

- Teamwork
- Time Management
- Communication
- Problem Solving
- Analytical Thinking
- Software Development Process



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CONCLUSION

The **Bank Management System Using Java** was successfully developed during the 40-day internship. The project automated essential banking operations such as account creation, secure login, deposits, withdrawals, money transfers, transaction history, ATM management, loan processing, fixed deposit calculations, and filebased data storage.

The internship provided practical exposure to software development and strengthened my understanding of Java programming, objectoriented design, debugging, testing, and documentation. The completed project fulfills the objectives established at the beginning of the internship and demonstrates the application of programming concepts in a real-world banking scenario.



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REFERENCES

The following resources were consulted during the internship:

Books

1. Herbert Schildt – *Java: The Complete Reference*
2. Cay S. Horstmann – *Core Java*
3. E. Balagurusamy – *Programming with Java*

Websites

- Oracle Java Documentation
- Eclipse IDE Documentation
- Java Tutorials
- W3Schools Java Tutorial
- GeeksforGeeks Java Section



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APPENDIX

Screenshots to Include

Adding some of the SS Of the Project :

- Home Screen
- Account Creation
- Login Screen
- Main Menu
- Deposit
- Withdraw
- Money Transfer
- Mini Statement
- ATM Services
- Loan Module
- Fixed Deposit
- Exit Screen

FINAL SUMMARY

The **Bank Management System Using Java** is a comprehensive academic project developed during a 35-day internship. It integrates multiple banking features into a single Java application, including account management, secure authentication, financial transactions, ATM services, loan calculations, and persistent file storage. The project demonstrates



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practical application of Java programming concepts such as Object-Oriented Programming, file handling, collections, exception handling, and modular design.

Overall Training Achievements

- Successfully completed a 35-day internship.
- Designed and implemented a complete banking application.
- Applied Java programming concepts to solve real-world problems.
- Improved debugging, testing, and documentation skills.
- Gained practical software development experience.
- Prepared a professional internship report documenting the complete development process.